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Patterns of field layer invertebrates in successional stages of managed boreal forest: Implications for the declining Capercaillie *Tetrao urogallus* L. population

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ABSTRACT

Capercaillie (*Tetrao urogallus L.*) populations in Finland have decreased markedly during past decades. One of the assumed reasons is the decreased quality of brood feeding grounds since current forest management transfers mature forests to younger successional stages. We studied how different types of managed Finnish forests offer resources for capercaillie broods by comparing the vegetation and invertebrate fauna of four successional stages in the two most common forest types in Finland, spruce dominated *Vaccinium myrtillus* (MT) and pine dominated *Vaccinium vitis-idaea* (VT) type. Forest age class had a significant effect on the cover of bilberry. There was a positive correlation between bilberry cover and the biomass of larvae and of all invertebrates (including all developmental stages) both at the stand and the sample level. Both forest type and age class significantly affected the biomass of larvae. Mature stands and young stands had the highest biomass of larvae in both MT and VT. In both forest site types the sapling stands hosted the smallest biomass of larvae. The results suggest that successional stages that follow clear cutting seriously lowers the food availability for capercaillie chicks and also reduces the shelter that field layer might provide. However, already the young stages that follow sapling phases seem to provide both bilberries and invertebrates so it is quite unlikely that the bilberry abundance alone could explain the dramatic decline of the capercaillie.

1. Introduction

Several long-term monitoring studies have shown that capercaillie (*Tetrao urogallus L.*) populations have declined remarkably in Finland recently. For example, annual winter bird counts that cover most of the southern parts of the county indicate that the reduction has been up to 80–90% since the early 1950s (Väisänen and Solonen, 1997; Fig. 1). Similar but somewhat less dramatic changes are reported on the basis of long-term game censuses where reduction has been estimated around 55–70% (Helle and Helle, 1991; Lindén et al., 1996). Several hypotheses aiming to explain these trends have been put forward (Rajala, 1962; Milonoff et al., 1993; Rätti et al., 1998; Porkert, 1999; Kurki, 1999; Helle et al., 2000; Lindén, 2002a). The decline is often assumed to be related to changes that forestry has caused. In Finland and the other Nordic countries the landscape and forest structure has faced fundamental changes since the beginning of modern forestry in 1950s (Esseen et al., 1992; Linder and Östlund, 1998; Kouki et al., 2001). Forests have been converted into managed single species and single cohort stands with increasing cover of clear-cuts and young plantations (Peltola, 2006). However, the exact mechanisms that transfer the impacts of forestry on capercaillie populations are not well understood.

There is ample indirect evidence that the decline in capercaillie populations has been caused by increased mortality in early life stages (Storch, 1994; Kurki et al., 2000). Increased chick mortality can be caused by a lack of optimal food, especially during the early phases of chick development, or increased predation pressure (for a hypothesis on causes for increased predation in managed forests, see Henttonen, 1989).

One of the direct effects of forest management and especially of the clear cutting harvest of trees is the reduction in the cover of bilberry (*Vaccinium myrtillus*) (Reinikainen et al., 2001; Uotila and Kouki, 2005). In Finland the bilberry cover has decreased over 50% since the 1950s (Reinikainen et al., 2001) although some of the decrease may be caused by changes in survey methods (Solantie, 2006). This coincides with the beginning of commercial forestry in Finland and also with the decline of capercaillie (Fig. 1). Bilberry suffers from altered light and water conditions after clear cutting and mechanical damage caused by harvesting machines (Atlegrim and Sjöberg, 1996a). Bilberry berries are one of the most important

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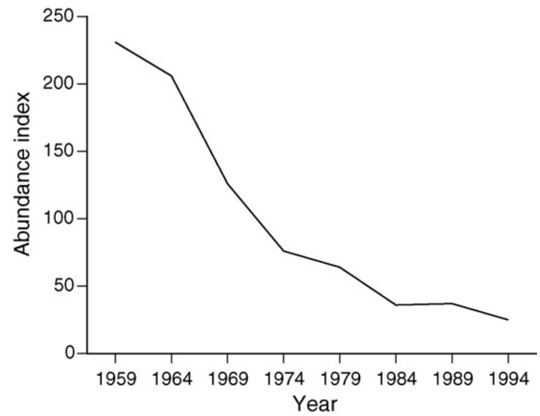

Fig. 1. The population trend of the Capercaillie in Finland from 1950s to 1990s. The data are based on annual winter bird censuses that cover whole country. The values are abundance indices based on 5-year averages (for the method of calculating the index, see Väisänen and Solonen, 1997. The original data are from Risto A. Väisänen, Finnish Museum of Natural History, and used here with a permission.).

food sources for chicks, and bilberry cover also gives small chicks shelter, especially from the avian predators (Wegge et al., 2005). Thus, the reduction of bilberry may have a major impact on the survival of the capercaillie chicks.

Bilberry is also an important host plant for many invertebrates, especially Lepidoptera larvae (e.g. Seppänen, 1970). Examples of common and widespread Lepidoptera feeding on bilberry include the geometrids *Eulithis populata* (Linnaeus, 1758) and *Chloroclysta citrata* (Linnaeus, 1761). Common sawflies include, for example, *Pristiphora carinata* (Hartig, 1837) and *Pristiphora mollis* (Hartig, 1837). During the first couple of weeks capercaillie chicks feed mainly on insect prey. In 1–2-week-old chicks, 80% of the stomach content consists of invertebrates that live on bilberry leaves (Kastdalen and Wegge, 1985; Spidsø and Stuen, 1988; Wegge and Kastdalen, 2008). Although Ledidoptera larvae form the main part of the early diet, invertebrates belonging to groups Formicidae, Hymenoptera, Coleoptera and Arainaea are also consumed (Sjöberg, 1996; Picozzi et al., 1999; Summers et al., 2004; Wegge and Kastdalen, 2008). After switching to plant food later on during their development bilberry is an important food source, and the chicks eat its berries, young leaves as well as shoots.

Consequently, a major mechanism by which forestry affects capercaillie populations might be through the effects on field layer vegetation – in particular by decreasing bilberry cover – that may affect availability of the protein-rich invertebrate food during the early chick development and also the berry-based food and shelter during the later development phases.

In this study we identify patterns of vegetation structure and invertebrate fauna across forest types and successional stages.

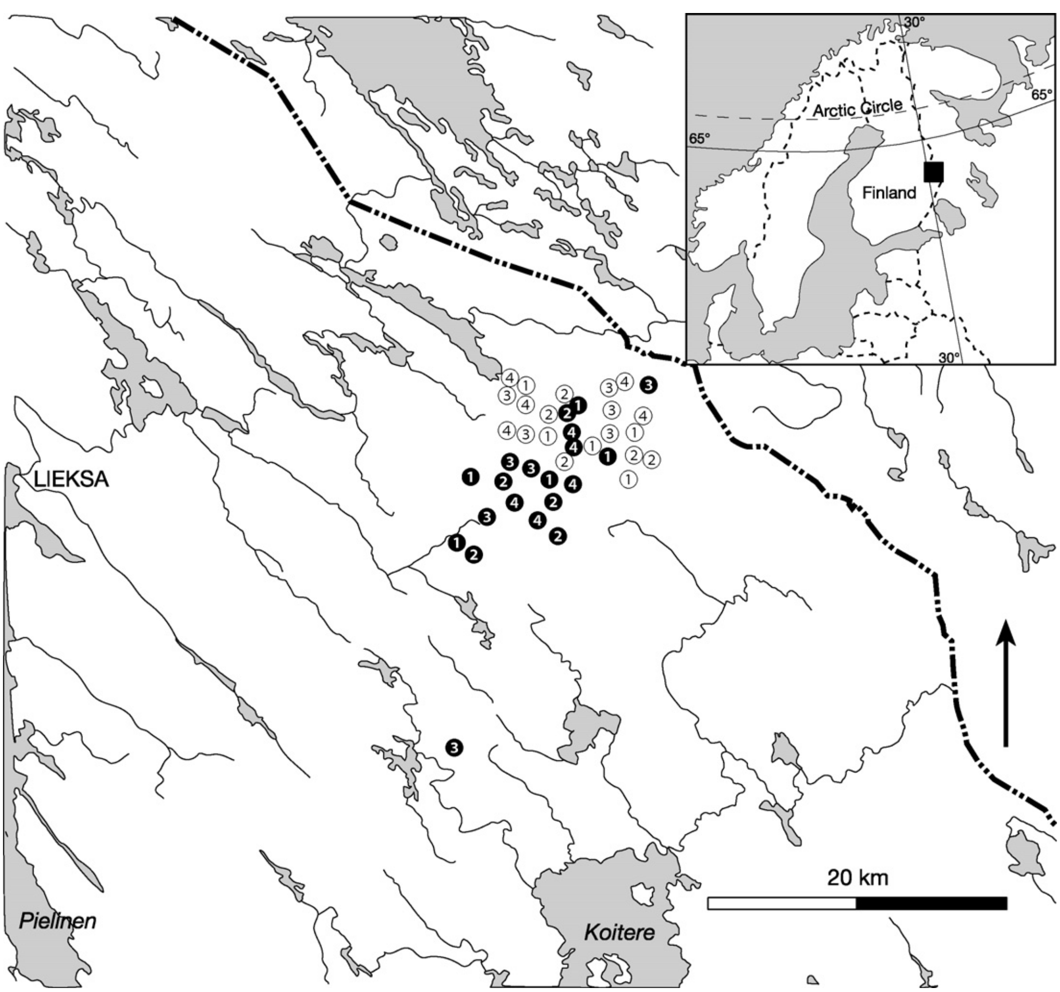

Fig. 2. Location of the study sites. Black symbols are spruce dominated MT forests, white symbols pine dominated VT forests. Numbers inside the symbols indicate the age class from sapling (1), young (2), middle-aged (3) to mature (4). Each type is replicated five times.

More specifically, we compare the vegetation and insect fauna of forests in four different successional stages from the two most common forest types in Finland, spruce dominated *Myrtillus* (MT) type and pine dominated *Vaccinium* (VT) type (for the Finnish forest classification, see Cajander, 1949). In particular, we aim at estimating what kind of habitat the different successional stages in different forest types offer, in terms of the available insect food, for forest grouse broods including the capercaillie.

Based on previous literature we set two hypotheses: (1) largest bilberry cover will be found from the oldest spruce dominated forests and (2) The largest biomass of insects will be found from study sites with the largest bilberry cover.

2. Material and methods

The research was conducted in Lieksa, Eastern Finland (63°10′N 30°40′E). The region belongs to the middle boreal vegetation zone. Forty study sites – each located in different forest stands – were located from two different forest site types, i.e. spruce dominated bilberry *Vaccinium myrtillus* type (MT) and pine dominated lingonberry *Vaccinium vitis-idaea* type (VT) (Fig. 2). In both forest types four different age classes were studied including: (1) sapling stands (selection criterium: average tree diameter <7 cm), (2) young stands (average diameter ≥7 cm and average height >7 m), (3) middle-aged stands (average diameter ≥17 cm) and (4) mature forest (ready for regeneration). Characteristics of the study sites are shown in Table 1. Consequently, each forest type × successional stage was replicated five times in the sampling design. The study sites were located in state-owned commercially managed forests and selected semi-randomly among the available ones. The stands represented the variation found in the managed stands in the region. For example, young stands were not yet thinned, but the middle-aged stands were thinned. Thinnings or tree densities were not used as a selection criterion because we wanted to include the typical variation that occurs in different successional classes.

On each study site five 1 m × 1 m sample squares were established and the abundance of mosses, lichens, graminoids, herbs and ericaceous shrubs was estimated visually with a projection cover (percentage scale 0–100%). We used these functional groups instead of species-level data, because our hypotheses relate to the cover of bilberry rather than the cover of every plant species. For species-level descriptions and analyses of the corresponding pine and spruce dominated managed forests in the study region, see Uotila et al. (2005) and Uotila and Kouki (2005).

Insects were sampled from the same 40 study sites with the sweep-net method. This included 40 sweeps at each site between 19 and 22 June 2006, which is during the period when newborn chicks are still small and feed on invertebrates in Eastern Finland. The weather conditions were almost identical during the sampling period including dry and sunny days with daytime temperatures of approximately 25 °C. The samples were collected during the

Table 1
Characteristics of the 40 study sites (A: 20 spruce dominated MT sites and B: 20 pine dominated VT sites). S.D. = standard deviation, –, no data available. Deciduous tree species that make only a marginal amount of the stands have been excluded from the table.

A. MT forests	Stand characteristic									
	Basal area (m^2/ha)		Stem density (stems/ha)		Height (m)		dbh (cm)		Age (years)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Age class and tree species										
Saplings	–	–								
Pine	–	–	100	141.5	1.3	1.8	0.8	0.7	10.5	5.6
Spruce	–	–	1580	286.4	1.4	1.6	1.2	1.6	10.0	5.1
Young										
Pine	2.0	2.2	70	67.1	13.0	4.7	17.3	5.1	41.7	7.8
Spruce	12.2	3.8	1000	430.2	12.2	2.2	15.8	3.1	39.8	4.9
Middle-aged										
Pine	1.6	2.2	–	–	16.0	2.2	19.5	3.4	68.0	40.2
Spruce	12.2	6.9	–	–	16.8	1.1	22.2	1.3	53.8	13.0
Mature										
Pine	4.0	4.5	–	–	21.0	3.4	30.3	8.5	153.0	16.1
Spruce	16.4	3.6	–	–	19.4	3.4	27.8	7.6	148.0	10.7

B. VT forests	Stand characteristic									
	Basal area (m^2/ha)		Stem density (stems/ha)		Height (m)		dbh (cm)		Age (years)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Age class and tree species										
Saplings										
Pine	–	–	3180	1217.3	1.7	1.3	1.9	1.6	10.4	4.0
Spruce	–	–	120	216.9	1.8	2.9	1.8	2.9	13.5	1.1
Seed trees (pine)	–	–	40	89.4	14.0	0.0	18.0	0.0	52.0	0.0
Young										
Pine	17.8	2.7	1100	308.1	11.4	2.0	13.8	0.9	39.0	9.8
Spruce	0.4	0.9	40	89.4	8.0	0.0	11.0	0.0	35.0	0.0
Middle-aged										
Pine	19.0	1.8	–	–	16.2	0.4	20.8	2.7	66.6	7.6
Mature										
Pine	18.6	3.6	–	–	19.2	1.6	27.2	3.4	112.4	20.8

daytime hours between 09.00 and 18.00. The size of the sweep-net was 30 cm in diameter.

Invertebrates in each of the 40 study sites were sampled during one visit. Several sites were sampled each day. Due to practical reasons, sites close to each other were sampled after each other, to keep the transfer times between sites reasonable. However, because the sites that represented different age classes were not located systematically close to each other (Fig. 2), there should not be any temporal bias that would jeopardize the estimates of within-group variability. In other words, forests from each age classes were sampled basically in any day and in any time between 09.00 and 18.00. The sweep samples were obtained by author JL so that she first walked roughly to the middle of the stand. Using that (haphazardly chosen) location as the centre point, sweep samples were obtained from a distance of 20 m to south, west, north and east directions of the centre point. Since she could not see the final sampling locations beforehand, we feel that this should guarantee that there is not a bias in the sampling procedure.

Each field sample contained 40 sweeps in each of the four locations (20 m from the centre to S, W, E, N) within a site, i.e. after sweeping 40 times with the net the invertebrates were collected from the net. Thus, there were four field samples from each site. To avoid pseudoreplication in the analyses, the averages of the four samples per site were used. This was required to obtain proper error terms and error d.f.'s in the analyses. The method of sweep netting has been shown to be a good survey method for invertebrates living in the field layer, especially when comparing differences in faunas. However, ground-dwelling arthropods like ants, which may occasionally be important in the chicks' diet, are underestimated by the method (Carpenter and Ford, 1936; Stuen and Spidsø, 1988).

The samples were frozen and later examined under a stereomicroscope. Invertebrates were identified into nine groups: Araneae, Heteroptera, Hymenoptera (adults), Diptera, Homoptera, Coleoptera (adults), Larvae (including Lepidoptera, Coleoptera, Hymenoptera larvae), Lepidoptera (adults), and others. After the identification the samples were dried in 105 °C for 24 h to measure the total biomass of each sample. Biomass was used in the analyses because the size of individuals in different groups varies greatly and therefore biomass gives an additional estimate of the energy and protein that is available for the chicks. In this respect, abundance might not be sufficient to evaluate the ecological importance of food availability (Saint-Germain et al., 2007). In total, the invertebrate data include 3058 individuals (Table 2).

Vegetation cover was analyzed with the parametric factorial ANOVA and the data were arcsine transformed because variables were percentages. For invertebrates the parametric ANOVA was used followed by Tukey's post hoc tests for those factors that were significant. The data were log transformed $\ln(x+1)$ before the analyses. The fit of the ANOVA model was evaluated from the error residual plots (observed vs. predicted) and the residual autocorrelations were tested with Durbin–Watson statistic. These did not indicate patterning of the residuals.

3. Results

3.1. Vegetation

Forest type significantly affected both the cover of bilberry ($p = 0.007$) and lingonberry ($p < 0.001$) (Table 3; Fig. 3). The cover of bilberry was also affected by age class ($p = 0.001$). There was no interaction between forest type and age class in bilberry cover but there was a significant interaction in lingonberry showing that the trend was different between MT and VT. In MT the cover was highest in sapling stands, then falling in young stands and rising again towards middle-aged and mature stands. In VT the lingonberry cover was lowest in sapling stands, then rising towards young and middle-aged stands and falling again in mature stands. Graminoids were significantly affected by both

Table 2
Numbers of invertebrates sampled from different taxonomic groups in (A) pine-dominated VT and (B) spruce-dominated MT forests. Numbers give the total number of individuals sampled from each forest age and forest type categories (=sum of individuals in five forest sites in each forest age class group).

A. VT pine forests	Forest age class				
	Sapling	Young	Middle-aged	Mature	Total
Larvae	8	80	42	31	161
Hymenoptera	15	25	9	18	67
Araneae	29	63	49	36	177
Coleoptera	14	4	11	15	44
Heteroptera	9	7	2	1	19
Diptera	2	4	6	11	23
Homoptera	49	40	2	14	105
Lepidoptera	0	4	1	0	5
Others	94	298	87	49	528
Total (VT)	220	525	209	175	1129

B. MT spruce forests	Forest age class				
	Sapling	Young	Middle-aged	Mature	Total
Larvae	13	84	60	93	250
Hymenoptera	17	49	30	29	125
Araneae	20	65	42	51	178
Coleoptera	7	1	5	8	21
Heteroptera	9	21	26	13	69
Diptera	16	90	65	48	219
Homoptera	90	20	31	16	157
Lepidoptera	0	3	4	5	12
Others	70	385	201	242	898
Total (MT)	242	718	464	505	1929
Total (MT + VT)	462	1243	673	680	3058

Table 3
Results of ANOVA for vegetation groups, showing the effects of forest type and age class on percentage vegetation cover. d.f., degrees of freedom; MS, mean square; *F*, *F*-ratio; Sig., statistical significance.

Source	d.f.	MS	*F*	Sig.
Mosses				
Forest type	1	0.03	0.77	0.386
Age class	3	0.10	2.57	0.072
Forest type × age class	3	0.07	1.97	0.139
Bilberry				
Forest type	1	0.37	8.38	0.007
Age class	3	0.30	6.82	0.001
Forest type × age class	3	0.02	0.47	0.702
Lingonberry				
Forest type	1	0.70	39.10	<0.001
Age class	3	0.01	0.54	0.658
Forest type × age class	3	0.13	7.19	0.001
Graminoids				
Forest type	1	0.18	7.38	0.011
Age class	3	0.10	4.20	0.013
Forest type × age class	3	0.10	4.27	0.012
Herbs				
Forest type	1	0.26	23.73	<0.001
Age class	3	0.01	0.96	0.423
Forest type × age class	3	0.05	4.82	0.007
Other dwarfs				
Forest type	1	0.46	18.42	<0.001
Age class	3	0.11	4.25	0.012
Forest type × age class	3	0.05	1.80	0.168

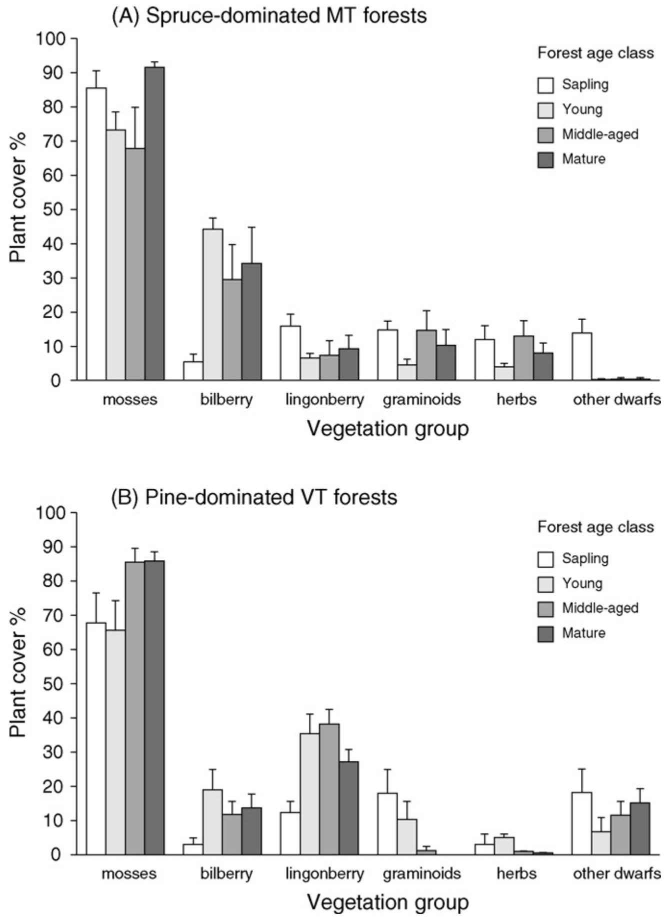

Fig. 3. Cover by mosses, bilberry, lingonberry, graminoids, herbs and other dwarfs in four different successional stages in MT (A) and VT (B) forests. Error bars indicate the standard errors of the means (S.E.; based on five replicate sites per bar).

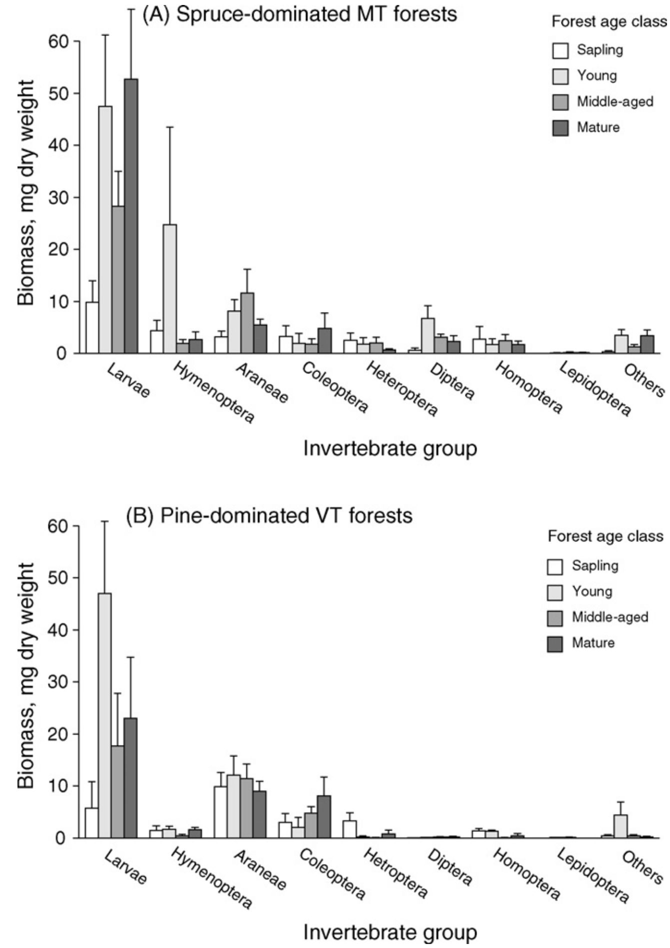

Fig. 4. The biomass (mg, dry weight, averaged over five replicates) of different invertebrate groups in four different successional stages in MT (A) and VT (B) forests. Error bars indicate the standard errors of the means (S.E.; based on five replicate sites per bar).

forest type ($p = 0.011$) and age class ($p = 0.013$) with a significant interaction. Herbs were significantly affected by forest type with significant interaction and other dwarfs by forest type and age class. Bilberry cover in sapling stands differed significantly from all other successional stages (young stands $p = 0.001$, middle-aged stands $p = 0.051$, mature stands $p = 0.016$). Between the other successional stages no significant differences were found.

The cover of mosses was highest in mature forests in both MT (91.52%) and VT (85.8%) (Fig. 3). Graminoids were most abundant in sapling stands (MT 14.8%, VT 17.92%). Highest percentage of lingonberry was found in young (35.4%) and middle-aged (38.2%) stands from VT. In MT and VT the percentage of bilberry cover was highest in young stands (MT 44.2%, VT 18.96%) followed by mature forest (MT 34.2%, VT 13.64%) and middle-aged stands (MT 29.48%, VT 11.76%). The bilberry cover was lowest in saplings stands (MT 5.44%, VT 3%).

3.2. Invertebrates

Larvae (including Lepidoptera, Hymenoptera and Coleoptera), Hymenoptera and Araneae had the highest biomass in MT while larvae, Araneae and Coleoptera had the highest biomass in VT (Fig. 4). When combined, the average total biomass of invertebrates was higher in MT compared to VT.

Both forest type ($p = 0.014$) and age class ($p < 0.001$) significantly affected the biomass of larvae (Table 4). For MT the highest biomass of larvae was found in mature forests (52.7 mg), followed by young stands (47.5 mg) and middle-aged stands (28.3 mg) and for VT young stands (47 mg), mature (23.0 mg), middle-aged stands (17.7 mg) (Fig. 4). In both forest site types the sapling stands hosted the least biomass of larvae (MT 9.9 mg, VT 5.7 mg). The differences in the biomass were significant when comparing sapling stands to young stands ($p < 0.001$) and mature stands ($p = 0.001$) and very close to significant between sapling stands and middle-aged stands ($p = 0.059$). The biomass of larvae in young, middle-aged and mature stands did not differ from each other significantly. For Araneae forest type was significant, the biomass for VT being higher than MT, and for Hymenoptera both forest type and age class significantly affected their biomass, which was higher in MT than VT. Others, including for example ticks and thrips, were also significantly affected by both forest type and age class. Only two groups (Diptera and others) showed significant interaction in the effects between stage and type. Thus, the patterns in invertebrate biomass can almost exclusively be explained by the separate main effects of forest type and successional stage.

According to Spearman correlation there was a positive association between bilberry cover and the biomass of larvae both at the stand (0.952, $p < 0.001$) and the sample level (0.371, $p = 0.018$) (Fig. 5) as well as the biomass of all invertebrates at stand (0.976 $p < 0.001$) and sample level (0.333, $p = 0.035$).

Table 4
Results of ANOVA for invertebrate groups, showing the effects of forest type and age class on invertebrate biomass. d.f., degrees of freedom; MS, mean square; *F*, *F*-ratio; Sig., statistical significance.

Source	d.f.	MS	*F*	Sig.
Larvae				
Forest type	1	6.23	6.83	0.014
Age class	3	9.11	9.99	<0.001
Forest type × age class	3	0.68	0.75	0.532
Hymenoptera				
Forest type	1	5.65	10.93	0.002
Age class	3	1.79	3.46	0.028
Forest type × age class	3	0.76	1.47	0.242
Araineae				
Forest type	1	1.99	4.67	0.038
Age class	3	0.50	1.17	0.335
Forest type × age class	3	0.35	0.81	0.496
Coleoptera				
Forest type	1	1.14	1.04	0.316
Age class	3	1.45	1.33	0.283
Forest type × age class	3	0.36	0.33	0.802
Heteroptera				
Forest type	1	1.06	2.14	0.153
Age class	3	0.87	1.77	0.173
Forest type × age class	3	0.43	0.88	0.464
Diptera				
Forest type	1	10.41	53.98	<0.001
Age class	3	1.12	5.81	0.003
Forest type × age class	3	0.94	4.85	0.007
Homoptera				
Forest type	1	1.32	3.34	0.077
Age class	3	0.20	0.51	0.676
Forest type × age class	3	0.54	1.37	0.27
Lepidoptera				
Forest type	1	0.03	1.95	0.172
Age class	3	0.03	1.85	0.158
Forest type × age class	3	0.01	0.62	0.605
Others				
Forest type	1	1.30	4.54	0.041
Age class	3	2.01	7.01	0.001
Forest type × age class	3	0.85	2.96	0.047

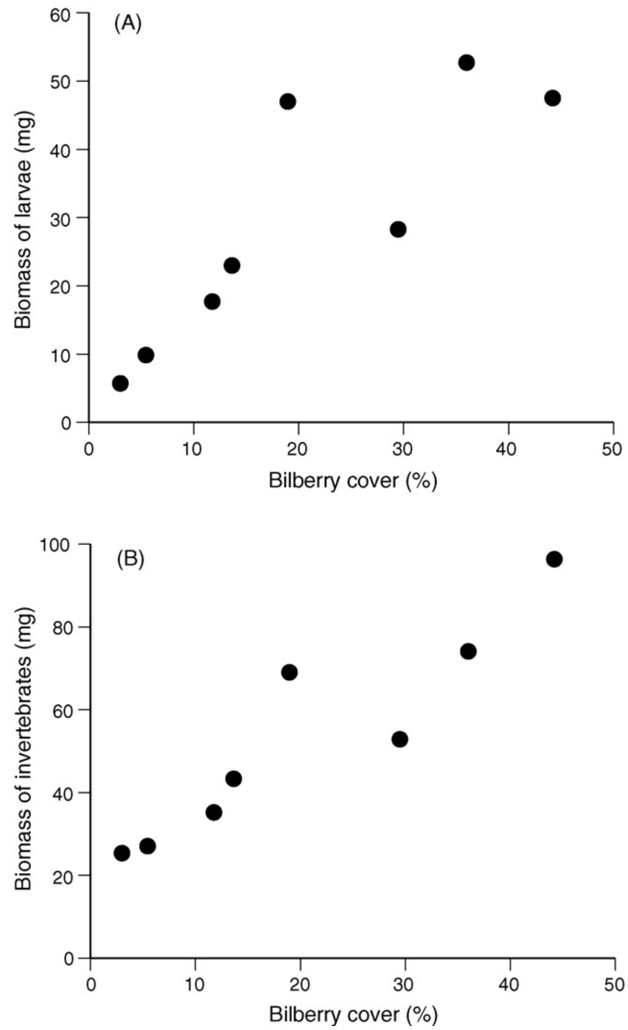

Fig. 5. Correlation between bilberry cover (%) and the biomass (mg, dry weight) of larvae (A) and invertebrates (B) at forest type level (four age classes in MT and VT).

4. Discussion

4.1. Food availability

Feeding and energy gain play an important role in the early survival of capercaillie chicks (Lindén, 1981, 2002b; Lindén et al., 1984) and therefore, studying the suitability of different types of forests as possible feeding grounds is highly important in order to understand survival of chicks and young birds. The type of invertebrate food eaten by capercaillie chicks seems to differ spatially and temporally (Rajala, 1960; Kastdalen and Wegge, 1985; Spidsø and Stuen, 1988) but these differences may depend on food availability. In our study larvae – mostly of Lepidopteran – constituted by far the largest group by biomass and based on previous research they are a very important source of energy for capercaillie chicks. In pristine boreal forest in Russia, capercaillie chicks clearly selected locations containing more lepidopteran larvae (Wegge et al., 2005). Differences in other insect groups were not as clear.

According to our results, sapling stands had the smallest biomass of invertebrate fauna compared to the rest of the age classes. Sapling stands also had the least amount of larvae in both MT and VT.

The abundance of invertebrates in clear-cut areas is significantly smaller than in old mature forest (Kastdalen and Wegge, 1985; Stuen and Spidsø, 1988). Bilberry suffers from direct sunlight and drought after clear cutting (Atlegrim and Sjöberg, 1996a). Higher light availability on clear-cut areas causes the phenol content of bilberry to rise and the water content to decrease. Therefore, its quality as a host plant for larvae is reduced and lower abundance of larvae is found from clear-cuts (Atlegrim and Sjöberg, 1996b).

Apart from sapling stands, all MT forests age classes had a relatively large amount of larvae. For Araneae the differences between age classes were not significant and for Hymenoptera the differences were caused by one large individual in one of the samples. Although the group ‘others’ was also significantly affected by both forest type and age class, this group consist of individuals that are probably too small in size to be of any importance to capercaillie chicks. Previously, a strong positive relationship has been found between the consumption of large sized larvae and the survival of capercaillie chicks (Picozzi et al., 1999).

Mature forest with lush ground vegetation and therefore high invertebrate abundance is often considered as the optimal habitat for capercaillie chicks (Storch, 1993; Sjöberg, 1996). The results of our study show, however, that also other forest age classes seem to be able to offer almost the same amount of food. The cover of bilberry may be of great importance and only the sapling stands (and possibly also all the recent clear-cuts) are clearly inferior for bilberry. In this study the amount of larvae and the cover of bilberry were correlated and the bilberry cover was in fact relatively high in young stands for both MT and VT, although the difference between mature stands was 10% and 5%, respectively. Increased canopy cover decreases the light that reaches the ground, which in turn reduces the cover of field layer including bilberry (Sjöberg, 1996). Although we lack detailed measurements of the canopy cover in our study sites, this may explain why mature stands had smaller bilberry cover than young stands. Young stands are also denser in shrub and field layer than middle-aged stands. Therefore, young stands may give capercaillie broods good cover and adequate availability of food.

Baines et al. (2004) found that capercaillie breeding success increased with bilberry cover, at least up to 15–20% cover. In our study, young MT forests and the older stages, with 29.5–44% of bilberry cover, seem suitable habitats for capercaillie broods. In VT forests on the other hand only young stands reached over 18% bilberry cover, other age classes remaining under 14%. Invertebrate biomass followed the trend of bilberry. Sapling stands with 5% bilberry cover are unlikely to offer enough food for forest grouse chicks because the biomass of invertebrates, especially larvae is significantly smaller than in other successional stages.

4.2. Habitat selection, predation, and the survival of chicks

Young stands (21–40 years) constitute the largest area of forest age classes in Finland (Peltola, 2006), and this is the situation in our research area in Lieksa as well (Räsänen et al., 2000). But if young stands are suitable brood areas, why have the capercaillie populations continued declining then? One possibility is that the landscape-level effects play a fundamental role. Fragmentation affects also young stands. They are often surrounded by clear cuttings and sapling stands and therefore predation affects the survival of chicks more than in habitats with continuous forest cover (Kurki and Lindén, 1995; Kurki et al., 2000, but see also Kouki and Väänänen, 2000).

The main cause of death for capercaillie chicks is predation (Kastdalen and Wegge, 1989; Wegge and Kastdalen, 2007). The probability that a predator finds a brood is likely to be affected by forest fragmentation or cover (Storaas et al., 1999; Kvasnes and Storaas, 2007). Several factors affect and modify the vulnerability of chicks to predation in fragmented areas. Weather conditions affect the survival of chicks but the impact of weather is connected to other factors such as food availability and shelter. During cold or rainy days chicks have to be warmed frequently by their mother and the time available for searching for food decreases. The chicks need additional energy for thermoregulation (Lindén et al., 1984). In optimal habitats where food is abundant, bad weather might not have such severe effects if the chicks can gain enough energy in a shorter time (Kastdalen and Wegge, 1989). If the field layer/bilberry cover is sufficient, capercaillie chicks do not have to move often between feeding places through areas of little shelter and, therefore, the possibility of being killed by a predator decreases. However, if optimal habitats are too small and a long distance apart, broods have to move through areas where they are exposed to predation.

Furthermore, when the percentage of suitable brood habitat decreases, the broods may concentrate on smaller forest patches where predators might easily locate them. Therefore, the benefit that predators gain from searching for broods increases exponentially with the decrease of brood habitat because searching becomes more effective with increased brood density (see also Sonerud, 1985; Kouki and Väänänen, 2000).

4.3. Conclusions

In conclusion, our results strongly suggest saplings stages that follow clear cutting seriously lowers the food availability for capercaillie chicks and most likely also reduces the shelter that field layer might provide. However, bilberry cover recovers within a few decades and already the young forest stages seem to provide both bilberries and invertebrates. Since clear-cut and sapling areas cover only a small proportion of a typical managed forest landscape, it is quite unlikely that the bilberry abundance alone could explain the dramatic decline in capercaillie. The landscape-level changes caused by forestry, such as fragmentation, may also cause increased predation pressure that could not be assessed in this study but is likely to modify the effects of food availability and shelter.

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